

V_{ud} extraction from EW corrections

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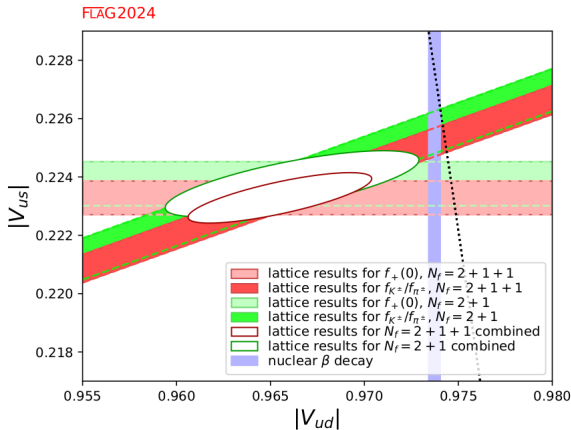
- ① CKM matrix
- ② V_{ud} from neutron decay
- ③ V_{ud} EFT framework
- ④ NLO EW corrections
- ⑤ NNLO EW corrections

- **CKM matrix** $\longrightarrow$ quark flavour mixing in weak interactions;
- First-row unitarity test

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1,$$

provides a sensitive probe of the SM;

- Current determinations of V_{ud} and V_{us} show a mild tension $\longrightarrow$ **Cabibbo Angle Anomaly**;
- Precision measurements of weak decays and accurate theoretical inputs are needed to clarify this tension;
- This talk:
 - Status of V_{ud} from superallowed β decays, pion decay and neutron decay;
 - Ongoing work on **NNLO EW corrections for neutron decay** and V_{ud} extraction.



V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays, FLAG2024.

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f (1 + 3g_A^2) (1 + \Delta_R^V)$$

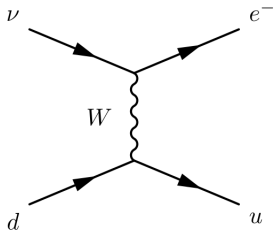
- f phase-space factor
- g_A axial coupling
- Δ_R^V universal electroweak radiative correction.

$$\Delta_R^V = 2.436(16)\%$$

$$|V_{ud}|_n = 0.97410(41)$$

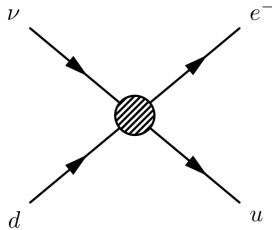
- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections;
- Reduced scale and scheme dependence;
- Improved CKM first-row unitarity tests;

$$d \rightarrow u e \bar{\nu}_e$$



Standard Model

$$m_W \gg k$$



Effective theory

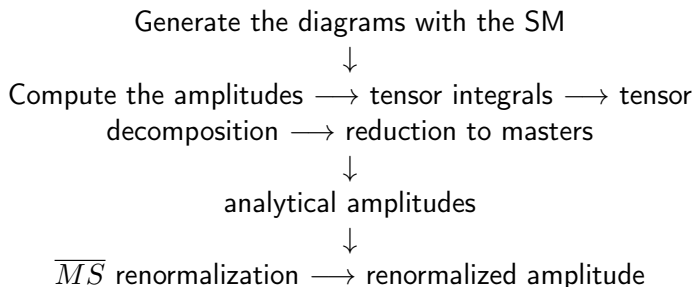
$$\mathcal{H}_{\text{eff}} = \frac{G_F}{\sqrt{2}} V_{ud} C_q(\mu) Q_q$$

$$Q_1 = (\bar{d}\gamma^\mu P_L u)(\bar{\nu}\gamma^\mu P_L e)$$

$$E_1 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\beta P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\beta P_L e) - (16 - a_q^1\epsilon - a_q^2\epsilon^2)Q_1$$

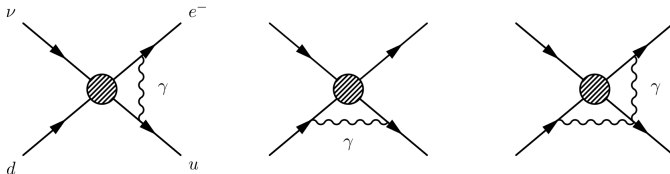
$$E_2 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\gamma^\sigma P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta\gamma_\sigma P_L e) - (256 - b_q^1\epsilon - b_q^2\epsilon^2)Q_1 + c_q^0 E_1$$

- d dimensions (vanish in 4 dimensions);
- scheme dependent;



Choice of kinematics: external momenta set to zero and all masses below the W mass are set to zero. Only masses present in our calculation will be m_W , m_Z , m_H , m_t .

1 loop EFT contribution for pion decay.



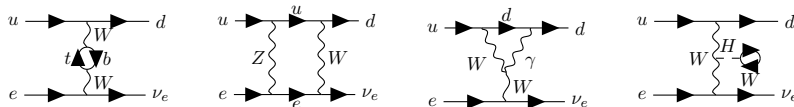
Exact decomposition of a propagator

$$\frac{1}{(k+p)^2 - m^2} = \frac{1}{k^2 - M^2} + \frac{-p^2 + 2k \cdot p - m^2 + M^2}{k^2 - M^2} \frac{1}{(k+p)^2 - m^2}$$

k and p stand for a linear combination of the internal moment and external momenta. The propagator mass is m and M is an introduced fictitious regulator.

Matching condition

$$\mathcal{A}_{\text{SM}} = \mathcal{A}_{\text{EFT}} = V_{ud} C_q Q_q$$



Representative one-loop EW diagrams contributing to the semi-leptonic process.

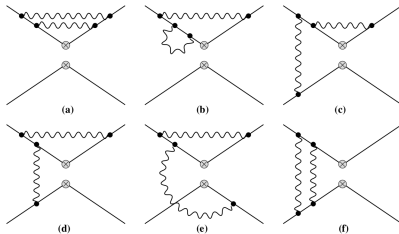
Wilson coefficient

$$C_q(\mu) = 1 + \frac{\alpha}{4\pi} C_q^{\text{NLO}} + \left(\frac{\alpha}{4\pi}\right)^2 C_q^{\text{NNLO}}$$

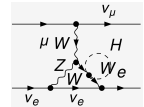
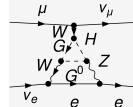
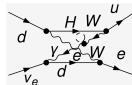
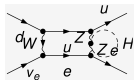
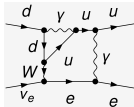
$$C_q^{\text{NLO}} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2 \ln\left(\frac{\mu^2}{M_Z^2}\right)$$

- Matching performed at the electroweak scale.
- Universal corrections cancel in the ratio of the quark decay normalized to muon decay: $\frac{C_q}{C_\mu}$.
- Only box-diagram contributions survive.

NNLO EW corrections



EFT NNLO EW corrections.



SM NNLO EW corrections.

- NNLO requires inclusion of boxes and penguins
- Tadpoles are included in the matching: gauge independent NNLO EW corrections and potentially substantial contribution in scheme conversion: $\overline{MS}$ to On-Shell.

WC depends on μ :

$$\mu \frac{d}{d\mu} C_i^f(\mu) = \sum_j \gamma_{ij}^f C_j^f(\mu), \quad (1)$$

depends on active number of flavours;

Anomalous dimension for muon decay (physical operator)

$$\gamma_\mu^{(0)} = 0, \quad \gamma_\mu^{(1)} = \left(\frac{2}{3} N_\ell + \frac{8}{27} n_c N_u + \frac{2}{27} n_c N_d \right) (4 - a_\mu^1), \quad (2)$$

$$\gamma_q^{(0)} = -4, \quad \gamma_q^{(1)} = \left(\frac{16}{9} N_\ell + \frac{64}{81} n_c N_u + \frac{16}{81} n_c N_d \right) (a_q^1 - 8). \quad (3)$$

Solution of Renormalization Group Equations for the Wilson Coefficient

$$C^{(f)}(\mu) = J_f(\mu) U_f(\mu) U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (4)$$

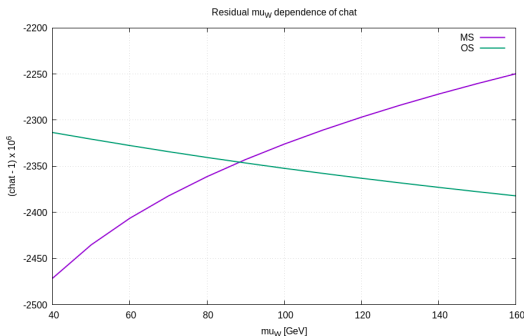
Evolution kernel at LO

$$U_f(\mu) = \left(\frac{\alpha^{(f)}(\mu)}{\alpha(M_Z)} \right)^{-\frac{\gamma^{(0)}}{2\beta^{(0)}}} \quad (5)$$

scheme independence

$$C = U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (6)$$

Wilson-Coefficient residual scale dependence



Scale dependence of the scale independent Wilson coefficient at NLO.

- Residual scale dependence at NLO induces an uncertainty $\delta C \simeq \pm 2 \times 10^{-4}$, which is significant for precision determinations of V_{ud} .
- NNLO corrections reduce the uncertainty to $\delta C < 2 \times 10^{-5}$, making perturbative uncertainties essentially negligible.

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.